

Experience Graph using Spatio-Temporal Scene Data for Replaying Mixed Reality Interaction

Seonji Kim*
KAIST UVR Lab

ABSTRACT

This study proposes an experience graph, a Mixed Reality (MR) experience replaying method for recording and playing spatially adaptive interaction in a mixed reality environment. Many MR immersive interactions are experienced in a space different from the original recorded space, so it is important to adaptively experience the interaction in the local space. As a solution to the collision and mismatch during the interactions in local space, there is research on scene generation or adaptive object placement, but the change of interactions dependent on space and the resulting experience transformation were not considered. This study structures spatial information and dependent interactions into an experience graph and proposes a spatial adaptive replay method that minimizes interaction deformation. Recorded interactions are evaluated quantitatively through loss rate and transformation rate, played interactions are evaluated qualitatively through user experiments, and finally, indicators important to the user experience are reflected in the experience graph. Utilizing the results of this study will contribute to experiencing immersive spatial MR where users actively record and share spatial experiences beyond the limitations of time and space.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Mixed Reality (MR) enables new experiences beyond the limits of real space. However, in order to experience such new experiences in an immersive way, users must be able to perform the interactions possible in a given Virtual Reality (VR) or Augmented Reality (AR) in real life. Real space still becomes an obstacle to various interactions such as walking, touching, and sitting in a virtual environment. These limitations are further emphasized when an interaction set in real space is experienced in another space. This study proposes an experience graph, a spatio-temporal scene graph for replaying shared MR experiences, and a method for replaying MR experiences to directly experience MR interactions beyond the limitations of space and time.

To convey experiences by replaying interactions made in real space, in the field of remote training, actions were recorded as they were or indirectly explained experiences intending to convey information remotely, but this was a sharing method that could not be directly experienced in other spaces. Research on creating new scenes to directly experience a given interaction was still a method that did not consider the real environment. Research on adaptively arranging interaction objects or avatars in a given space did not consider the transformation of the experience due to adaptive arrangement. This study aims to directly experience the original interaction across time and space by minimizing the degree of deformation of the experi-

ence by adaptively playing the recorded interaction in the original space in another space.

The experience graph, which records interactions based on a given space and adaptively replays them in other spaces, records interactions dependent on the space based on geometric and semantic information. Additionally, appropriate event-based recording and replay frequencies will be identified through user evaluation. Finally, we would like to develop an indicator to measure the loss rate and transformation rate of shared experience through the experience graph and evaluate the usefulness of the proposed method based on this. Through this, the goal is to complete the optimal experience recording method that minimizes loss and transformation of experience and the experience graph data structure for this. This study seeks to find answers to three major research questions.

1. What is the appropriate unit for structuring given space for spatially adaptive interaction recording and replay?
2. What is the appropriate event recording frequency for adaptively replaying interactions in space?
3. How to play spatially adaptive interactions to minimize changes in user experience?

2 RELATED WORKS

Remote collaboration methods have been studied for learning knowledge or skills by allowing remote users to directly view or experience interactions performed by users [3]. In particular, most research has been done on remote systems for communicating with each other when one user has different roles, such as a task instructor and another user has different views or different information, such as a task performer. However, in many cases, the goal was to indirectly confirm each other's real-time experience through visual cues, voices, and graphic interfaces. They attempted to convey accurate knowledge by recording one user's interaction as is and replaying it for another user to check, given the same object or environment. However, in cases where the real environment differs from the recorded environment, it was played regardless of the real environment in video or VR. Therefore, this study allows users to perform an immersive experience by adaptively playing recorded interactions even when the replay background space is different from the recording background space.

Research has been conducted to create new scenes or spaces that can perform the same or similar interactions as the given interactions [4]. By finding furniture objects that have the affordance to perform a given interaction and placing them in appropriate locations, we have created a natural and purposeful space. Additionally, research has been done to extract interactions from videos or other scenes and create other similar scenes. The final scene created is a likely scene based on the dataset, and it is possible to indirectly experience the targeted interaction in the scene through VR. However, because the created space is unrelated to the information of the current real space, it is impossible to directly experience the interaction in an immersive way in real space, so this study studies the experience graph for this purpose.

To enable the same interaction in space, research has been conducted to adaptively place interaction objects or avatars in different

*e-mail: seonji.kim@kaist.ac.kr

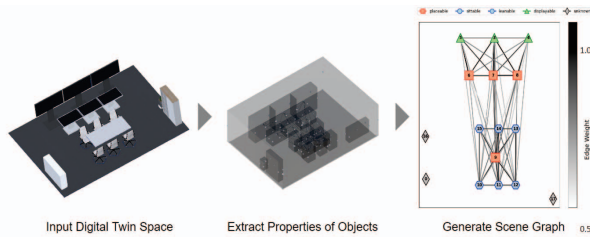


Figure 1: Scene graph using both geometric and semantic information of objects from progressed research.

spaces. It was possible to directly experience the target interaction or story in real space by arranging the interaction target object according to the shape and size of the given real space within the space, or by positioning the avatar considering the relative position of the interaction target object. These studies have been evaluated as wildfire experiments with objectives such as whether the deployment results are natural or the experience is enjoyable [2, 5]. However, the impact on users of changes in experience caused by variations in space-dependent interactions was not considered. This study aims to experience replayed interactions in other spaces as close to existing experiences.

3 RESEARCH PROGRESS

The scene graph representation method considering user interactivity has been studied. To record and play spatially adaptive interactions, common elements that enable the same interaction must be found in different spaces through structured spatial information. In the studied scene graph, object nodes were formed by considering both geometric and semantic information, including the affordance of each object, and the relationships between objects that can be used for interaction were calculated and expressed as edges (Fig. 1). The formed scene graph contains a group of objects that can be used together to view the space at a higher level rather than interacting with a single object. Additionally, common object groups were detected through comparison between scene graphs. To prove the effectiveness of the studied scene graph, an evaluation was conducted to find similar clusters in the generated spatial dataset with multiple clusters. When performing one-to-one matching between different spaces, the probability of interaction increased significantly when discovered similar clusters were taken into account, without significant changes in area loss compared to when they were not taken into account.

4 RESEARCH PLAN

In the future, the scene graph developed to find similar multiple cluster pairs will ultimately lead to the creation of an experience graph for experience replaying by considering movement between clusters in space. Afterward, effective methods of recording and playing interactions dependent on structured spatial information will be studied. In the case of existing remote training systems, the real-time location information of each object of interaction was recorded as is to deliver accurate information. However, continuous data accumulation without considering the interaction may not be an effective recording method. Therefore, a method of recording the experience by adjusting the update frequency according to the occurrence of interaction events will be studied, taking into account the modification of the experience.

The usefulness of the finally completed spatio-temporal scene graph and the experience replaying system utilizing it will be evaluated to preserve the experience. For this purpose, the loss rate and transformation rate of experience will be developed and used as quantitative indicators. The loss rate of the experience will be calculated by comparing the missing nodes and edges in the replayed

scene graph to the original scene graph. The transformation rate of the experience will be calculated through the frame-by-frame heterogeneity rate of recorded and played interactions. Additionally, as a qualitative evaluation, a questionnaire will be collected asking about the similarity between the original experience and the modified experience. Evaluations will be conducted using the Matterport 3D dataset, a large-scale real-space dataset [1]. By utilizing both qualitative and quantitative evaluations, the usefulness of the proposed MR experience replaying system in preserving experience can be evaluated from various angles.

5 DISCUSSION

This study proposes a method to record interactions in a specific space and adaptively replay them in another space to share experiences in an MR environment. The goal is to structure information in space through an experience graph, record the continuous interactions that depend on it, and play with minimal interaction transformation. To this end, the basis for interaction recording will be laid by structuring spatial information into not only object clusters but also paths. User experiments are then conducted to determine the appropriate frequency of event-based recording and play as successive interactions occur. Finally, the experience graph and MR experiences replaying method will be completed, reflecting the discovered indicators from evaluations. To that end, the questions I would like to discuss are as follows:

1. Is it appropriate to compare the original experience graph and the replayed experience graph as a quantitative evaluation through the loss rate?
2. Is it appropriate to compare original experience and replayed experience as a quantitative evaluation through the transformation rate?
3. Is it appropriate to use a questionnaire asking the similarity between original experience and replayed experience as a qualitative evaluation?

ACKNOWLEDGMENTS

This research was supported by National Research Council of Science and Technology (NST) funded by the Ministry of Science and ICT (MSIT), Republic of Korea (Grant No. CRC 21011) and the MSIT (Ministry of Science and ICT), Korea, under the Graduate School of Metaverse Convergence support program (IITP-2024-RS2022-00156435) supervised by the IITP (Institute for Information & Communications Technology Planning & Evaluation).

REFERENCES

- [1] A. Chang, A. Dai, T. Funkhouser, M. Halber, M. Niessner, M. Savva, S. Song, A. Zeng, and Y. Zhang. Matterport3d: Learning from rgb-d data in indoor environments. *International Conference on 3D Vision (3DV)*, 2017.
- [2] C. Li and L.-F. Yu. Generating activity snippets by learning human-scene interactions. *ACM Transactions on Graphics (TOG)*, 42(4):1–15, 2023.
- [3] Z. Liu, Z. Zhu, E. Jiang, F. Huang, A. M. Villanueva, X. Qian, T. Wang, and K. Ramani. Instrumentar: Auto-generation of augmented reality tutorials for operating digital instruments through recording embodied demonstration. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pp. 1–17, 2023.
- [4] R. Ma, A. G. Patil, M. Fisher, M. Li, S. Pirk, B.-S. Hua, S.-K. Yeung, X. Tong, L. Guibas, and H. Zhang. Language-driven synthesis of 3d scenes from scene databases. *ACM Transactions on Graphics (TOG)*, 37(6):1–16, 2018.
- [5] S. Marwecki and P. Baudisch. Scenograph: Fitting real-walking vr experiences into various tracking volumes. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology*, pp. 511–520, 2018.